Power BI Road Safety Dashboard - Complete Step-by-Step Implementation

PHASE 1: DATA PREPARATION

Step 1: Import Data

1. Open Power BI Desktop
2. Click **Get Data** → **Excel** (or your data source)
3. Select your accident data file
4. In Power Query Editor:
   - Check data types (Date should be Date, Time should be Time, etc.)
   - Remove any completely empty rows
   - Click **Close & Apply**

Step 2: Create Date Table

1. Go to **Modeling** tab → **New Table**
2. Enter this DAX:

```dax
DateTable =
ADDCOLUMNS(
    CALENDAR(MIN('Accident_Data'[Accident_Date]), MAX('Accident_Data'[Accident_Date])),
    "Year", YEAR([Date]),
    "Month", MONTH([Date]),
    "MonthName", FORMAT([Date], "MMMM"),
    "DayOfWeek", WEEKDAY([Date], 2),
    "DayName", FORMAT([Date], "DDDD"),
    "IsWeekend", IF(WEEKDAY([Date], 2) >= 6, "Weekend", "Weekday")
)
```

3. Mark as Date Table: Select table → **Table Tools** → **Mark as Date Table**

Step 3: Create Essential Calculated Columns

Go to **Data** view, select your Accident_Data table:

Hour Column:

```dax
```

```
Hour = HOUR('Accident_Data'[Time])
```

Time Period Column:

```dax
Time_Period =
SWITCH(
    TRUE(),
    'Accident_Data'[Hour] >= 6 && 'Accident_Data'[Hour] < 12, "Morning (6AM-12PM)",
    'Accident_Data'[Hour] >= 12 && 'Accident_Data'[Hour] < 18, "Afternoon (12PM-6PM)",
    'Accident_Data'[Hour] >= 18 && 'Accident_Data'[Hour] < 22, "Evening (6PM-10PM)",
    "Night (10PM-6AM)"
)
```

Severity Score Column:

```dax
Severity_Score =
SWITCH(
    'Accident_Data'[Accident_Severity],
    "Slight", 1,
    "Serious", 3,
    "Fatal", 5,
    0
)
```

Step 4: Create Relationships

1. Go to **Model** view
2. Drag from **DateTable[Date]** to **Accident_Data[Accident_Date]**
3. Ensure relationship is active (solid line)

Step 5: Create Basic Measures

Go to **Data** view → **New Measure**:

```dax
```

```
Total Accidents = COUNTROWS('Accident_Data')

Serious Accidents = CALCULATE(COUNTROWS('Accident_Data'), 'Accident_Data'[Accident_Severity] = "Serious")

Slight Accidents = CALCULATE(COUNTROWS('Accident_Data'), 'Accident_Data'[Accident_Severity] = "Slight")

Fatal Accidents = CALCULATE(COUNTROWS('Accident_Data'), 'Accident_Data'[Accident_Severity] = "Fatal")
```

PHASE 2: KPI IMPLEMENTATION

KPI 1: Accident Severity Distribution

Step 1: Create the Visual

1. Go to **Report** view
2. Click **Donut Chart** from visualizations
3. Drag **Accident_Severity** to **Legend**
4. Drag **Total Accidents** to **Values**

Step 2: Format the Visual

1. Click **Format Visual** (paint brush icon)
2. **Title**: Turn on, rename to "Accident Severity Distribution"
3. **Data Labels**: Turn on, show percentage
4. **Colors**:
   - Fatal: Red (#FF0000)
   - Serious: Orange (#FF8C00)
   - Slight: Yellow (#FFD700)

Step 3: Add Data Labels

- In **Data Labels** → **Values** → Select "Category, percentage of total"

KPI 2: Number of Accidents Over Time

Step 1: Create Line Chart

1. Add **Line Chart** visual
2. **X-axis**: DateTable[Date]
3. **Y-axis**: Total Accidents
4. **Legend**: Accident_Severity (optional)

Step 2: Format Timeline

1. **X-axis** → **Type**: Continuous
2. **Title**: "Daily Accident Trends"
3. Add trend line: **Analytics** → **Trend Line** → Add

Step 3: Create Monthly Summary (if needed)

1. Create measure:

```dax
Monthly Accidents = CALCULATE([Total Accidents], DATESMTD(DateTable[Date]))
```

2. Use DateTable[MonthName] on X-axis instead

KPI 3: Distribution of Accidents by Day of Week

Step 1: Create Column Chart

1. Add **Clustered Column Chart**
2. **X-axis**: Day_of_Week
3. **Y-axis**: Total Accidents

Step 2: Sort by Day Order

1. Select Day_of_Week column in Data view
2. **Column Tools** → **Sort By Column** → Create numeric day order column first:

```dax
Day_Order =
SWITCH(
    'Accident_Data'[Day_of_Week],
    "Monday", 1,
    "Tuesday", 2,
    "Wednesday", 3,
    "Thursday", 4,
    "Friday", 5,
    "Saturday", 6,
    "Sunday", 7
)
```

3. Sort Day_of_Week by Day_Order

Step 3: Add Insights

- Create measure for weekend vs weekday:

```dax
Weekend Accidents = CALCULATE([Total Accidents], DateTable[IsWeekend] = "Weekend")
Weekday Accidents = CALCULATE([Total Accidents], DateTable[IsWeekend] = "Weekday")
```

KPI 4: Top Junction Controls Contributing to Accidents

Step 1: Create Horizontal Bar Chart

1. Add **Clustered Bar Chart**
2. **Y-axis**: Junction_Control
3. **X-axis**: Total Accidents

Step 2: Filter Top Results

1. Click visual → **Filters** pane
2. **Filters on this visual** → Add Junction_Control
3. **Filter Type**: Top N
4. **Show Items**: Top 5
5. **By Value**: Total Accidents

Step 3: Create Risk Score

```dax
Junction Risk Score =
CALCULATE(
    AVERAGE('Accident_Data'[Severity_Score]),
    ALLEXCEPT('Accident_Data', 'Accident_Data'[Junction_Control])
)
```

KPI 5: Weather Conditions vs. Accident Severity

Step 1: Create Matrix Visual

1. Add **Matrix** visual
2. **Rows**: Weather_Conditions
3. **Columns**: Accident_Severity
4. **Values**: Total Accidents

Step 2: Add Conditional Formatting

1. Click **Values → Total Accidents → Conditional Formatting → Background Color**
2. **Format Style**: Gradient
3. **Minimum**: White
4. **Maximum**: Red

Step 3: Create Weather Risk Measure

```dax
Weather Severity Index =
CALCULATE(
    AVERAGE('Accident_Data'[Severity_Score]),
    ALLEXCEPT('Accident_Data', 'Accident_Data'[Weather_Conditions])
)
```

KPI 6: Road Surface Conditions vs. Accident Severity

Step 1: Create Clustered Bar Chart

1. Add **Clustered Bar Chart**
2. **Y-axis**: Road_Surface_Conditions
3. **X-axis**: Total Accidents
4. **Legend**: Accident_Severity

Step 2: Create Road Risk Score

```dax
Road Surface Risk =
CALCULATE(
    AVERAGE('Accident_Data'[Severity_Score]),
    ALLEXCEPT('Accident_Data', 'Accident_Data'[Road_Surface_Conditions])
)
```

Step 3: Alternative Heatmap View

- Use Matrix with Road_Surface_Conditions vs Accident_Severity
- Apply same conditional formatting as weather analysis

KPI 7: Peak Hour Accident Analysis (Additional KPI)

Step 1: Create 24-Hour Heatmap

1. Add **Matrix** visual
2. **Rows**: Hour
3. **Values**: Total Accidents

Step 2: Format as Heatmap

1. **Conditional Formatting** → **Background Color** → Gradient
2. **Style**: Red-Yellow-Green (reverse)
3. **Data Labels**: Show values

Step 3: Create Time Period Analysis

1. Add **Donut Chart**
2. **Legend**: Time_Period
3. **Values**: Total Accidents

Step 4: Peak Hour Measure

```dax
Peak Hour =
VAR MaxAccidents = MAXX(ALL('Accident_Data'[Hour]), [Total Accidents])
RETURN CALCULATE(MAX('Accident_Data'[Hour]), [Total Accidents] = MaxAccidents)
```

KPI 8: Geographic Risk Hotspots (Additional KPI)

Step 1: Create Map Visual

1. Add **Map** visual
2. **Location**: Latitude and Longitude (both fields)
3. **Size**: Number_of_Casualties
4. **Legend**: Accident_Severity

Step 2: Create Cluster Analysis

```dax
Location Risk Score =
CALCULATE(
    AVERAGE('Accident_Data'[Severity_Score]) * COUNT('Accident_Data'[Accident_Index]),
    ALLEXCEPT('Accident_Data', 'Accident_Data'[Latitude], 'Accident_Data'[Longitude])
)
```

Step 3: Add Address Details

- Create tooltip page with accident details
- **Page Settings** → **Tooltip** → Turn on

PHASE 3: DASHBOARD LAYOUT

Page 1: Executive Overview

Layout Structure:

```
[KPI Cards Row]    [Total] [Peak Day] [Peak Hour] [Risk Score]
[Main Charts]      [Severity Donut]   [Day of Week Bar]
[Secondary]        [Junction Analysis] [Time Trend Line]
```

Step 1: Add KPI Cards

1. Add 4 **Card** visuals
2. Configure each:
   - Total Accidents: `[Total Accidents]`
   - Peak Day: `CALCULATE(MAX(DateTable[DayName]), [Total Accidents] = MAXX(ALL(DateTable[DayName]), [Total Accidents]))`
   - Peak Hour: `[Peak Hour]`
   - Data Quality: `(1 - DIVIDE(COUNTROWS(FILTER('Accident_Data', 'Accident_Data'[Junction_Control] = "Data missing or out of range")), [Total Accidents])) * 100`

Step 2: Add Slicers

- Date Range Slicer (top of page)
- Weather Conditions (side panel)
- Vehicle Type (side panel)

Page 2: Temporal Analysis

Focus: Time-based patterns

- 24-hour heatmap (large, center)
- Day of week analysis (top)
- Weekend vs weekday comparison (bottom)
- Time period breakdown (side)

Page 3: Environmental Factors

Focus: Weather and road conditions

- Weather vs severity matrix (main)
- Road surface analysis (secondary)
- Light conditions impact
- Combined risk assessment

Page 4: Geographic & Infrastructure

Focus: Location and infrastructure

- Map with accident hotspots (main)
- Junction control analysis
- Road type breakdown
- Speed limit correlation

PHASE 4: ADVANCED FEATURES

Interactive Features

Step 1: Cross-Filtering

1. Select all visuals on a page
2. **Format** → **Edit Interactions**
3. Set appropriate filter/highlight relationships

Step 2: Drill-Through

1. Create new page "Junction Details"
2. Add Junction_Control to **Drill-through** filters
3. Add detailed visuals for selected junction
4. Right-click any junction on main page → **Drill Through**

Step 3: Bookmarks for Navigation

1. **View** → **Bookmarks Pane**
2. Create bookmarks for different views
3. Add buttons for navigation

Performance Optimization

Step 1: Optimize Measures

```dax
```

```
// Use SUMX instead of individual calculations where possible
Total Casualties = SUMX('Accident_Data', 'Accident_Data'[Number_of_Casualties])

// Use variables in complex measures
Risk Assessment =
VAR TotalAcc = [Total Accidents]
VAR AvgSeverity = AVERAGE('Accident_Data'[Severity_Score])
RETURN TotalAcc * AvgSeverity
```

Step 2: Reduce Data Model Size

- Remove unused columns
- Create summary tables for large datasets
- Use composite models if needed

PHASE 5: TESTING AND DEPLOYMENT

Testing Checklist

- [ ] All visuals load correctly
- [ ] Filters work across pages
- [ ] DAX measures return expected results
- [ ] Mobile layout is functional
- [ ] Performance is acceptable (<3 seconds load)

Publishing Steps

1. **File → Publish → Publish to Power BI**
2. Choose workspace
3. Configure data refresh schedule
4. Set up security (if needed)
5. Share with stakeholders

Training Materials

Create documentation covering:

- How to use slicers and filters
- Understanding each KPI
- Drill-through functionality
- Mobile access

TROUBLESHOOTING COMMON ISSUES

Data Issues

- **Missing values**: Filter them out or replace with "Unknown"
- **Date formatting**: Ensure consistent date format
- **Performance**: Use SUMMARIZE for large calculations

Visual Issues

- **Sorting**: Always create sort columns for custom orders
- **Colors**: Use consistent color scheme across all visuals
- **Labels**: Ensure readability with proper font sizes

DAX Issues

- **Context**: Use CALCULATE to modify filter context
- **Performance**: Avoid calculated columns for large tables
- **Testing**: Use DAX Studio for complex measure testing

This complete implementation guide will give you a professional, comprehensive road safety dashboard with all 8 KPIs and advanced analytics capabilities!